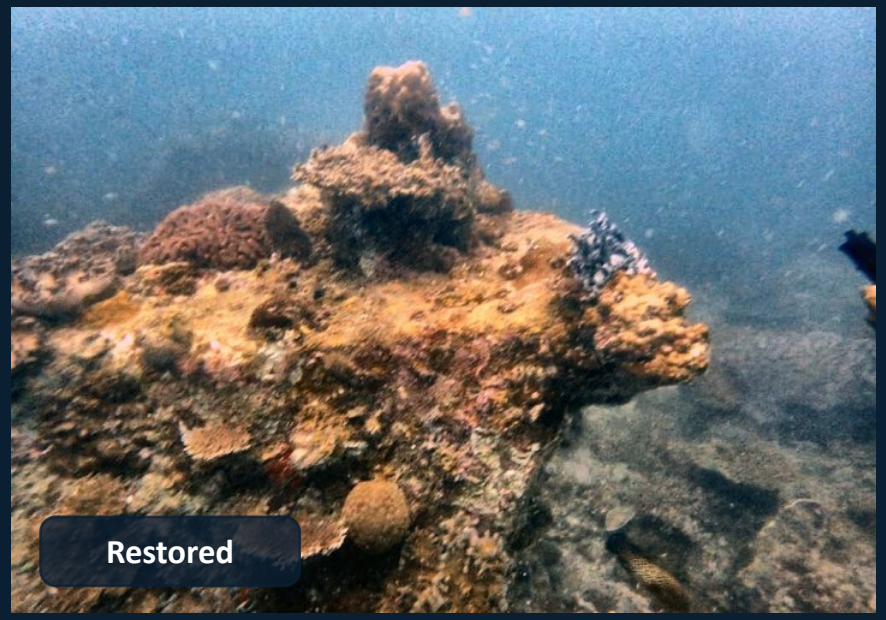
REVIEW II · BCSE403L DIGITAL IMAGE PROCESSING

Hybrid underwater image restoration

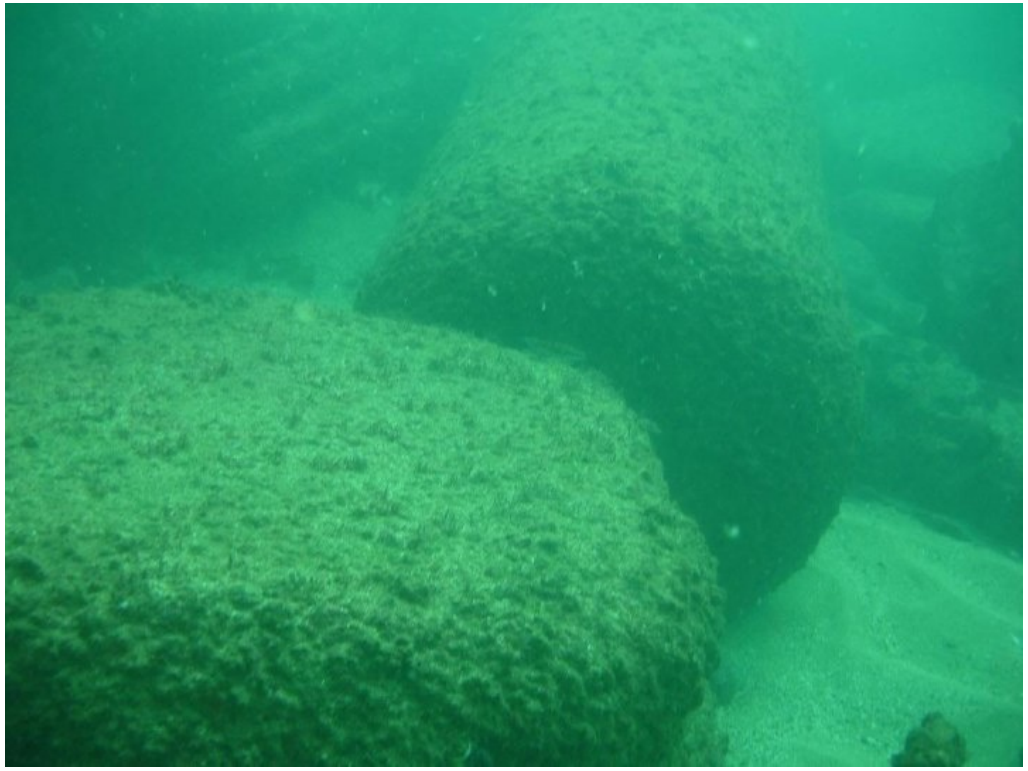
Adaptive colour correction, CLAHE and dehazing, tuned to the type of water in each image

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VIT School of Computer Science and Engineering · Fall 2026–27





Why underwater photos look wrong



Raw UIEB image from green, turbid water

Colour cast

Water absorbs red light first, then green. Images turn blue-green and lose their reds.

Haze

Suspended particles scatter light toward the camera, adding a veil that thickens with distance.

Low light and flat contrast

Light fades with depth and scattering flattens tones, hiding texture and detail.

Our goal and what is new

Restore underwater images with classical image processing only, adapt every stage to the type of water, and prove each gain on images never used for tuning.

MODULE 5

Water-type routing + red compensation

Classifies each image as clear blue, green turbid, deep low-light or hazy bright, then sets how strongly every stage acts, including how much red to restore.

MODULE 7

Per-image fusion weight

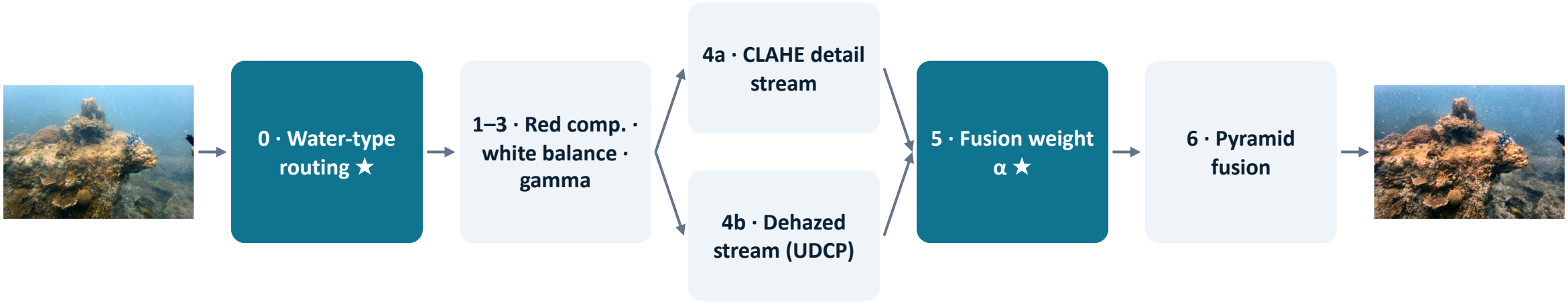
Predicts how to blend a detail stream with a dehazed stream for each image, from 26 no-reference image statistics.

MODULE 9

Downstream validation

Checks whether restoration helps computer vision, not just appearance: ORB keypoints, matches and edge density.

The pipeline at a glance



★ Novel contribution

Every stage on one held-out image



Raw input · classified clear blue



1 · Red compensation (s = 0.75)



2 · White balance (cap 1.5×)



3 · Gamma (γ = 1.14)



4a · CLAHE detail stream (clip 1.99)



4b · Dehazed stream (ω = 0.5)



5–6 · Fused result (α = 0.49)



UIEB reference

This image: PSNR 17.51 dB, SSIM 0.875 (CLAHE-only: 18.55 dB, 0.831)

Step 0 · Routing: four water types, two profiles

Deep low-light

mean brightness < 0.30

Green turbid

else if green dominance > 8 and haze > 0.35

Hazy bright

else if haze > 0.45 and saturation < 0.35

Clear blue

otherwise · 72% of UIEB

Profile	Red s	WB cap	Gamma ×	CLAHE ×	ω
Global (clear blue, green turbid and deep low-light)	0.75	1.5	1.05	0.6	0.5
Hazy bright	0.75	1.5	1.2	1	0.6

A type keeps its own values only if they beat the single global profile on its calibration images (paired Wilcoxon, $p < 0.05$). Clear blue, green turbid and deep low-light kept the global stage-1 values.

All values were chosen on the 30% calibration split only.

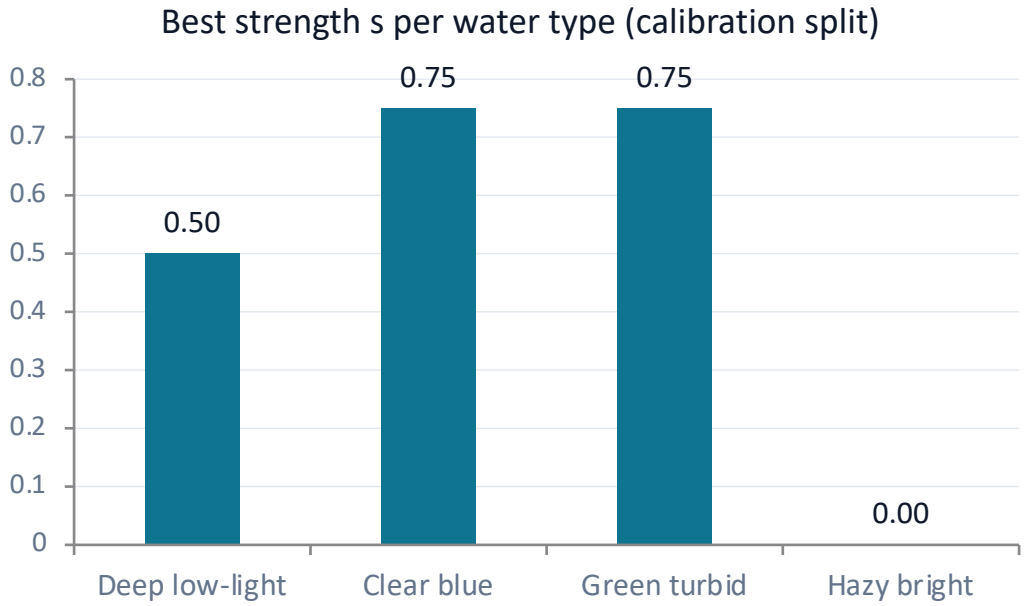
Step 1 · Restoring the red light water absorbs

$$R' = R + s \cdot (\text{mean } G - \text{mean } R) \cdot (1 - R) \cdot G$$

- Borrows red from the green channel where green is strong
- (1 − R) leaves pixels that are already red untouched
- Searched per water type; none beat one global strength (s = 0.75) significantly
- Applied before white balance, so balance has real red to work with (Ancuti et al., 2018)

+0.52 dB

PSNR on held-out images (95% CI 0.41–0.63), plus +0.024 SSIM — the largest single gain in the project



Steps 2–4 · Correct the colour, then build two streams

2

White balance

Gray-world gains make the average colour neutral; each channel at most doubled or halved.

3

Adaptive gamma

A gamma curve sets overall brightness (here γ = 1.14).

4

Two streams

The corrected image is processed two complementary ways, then fused.



4a · CLAHE on lightness (L*)

Equalises contrast tile by tile on L* only, so colours don't shift.

+ recovers texture and detail
− greyish and noisy on its own



4b · Underwater dark channel prior

Estimates haze per pixel from the G/B dark channel, then removes it.

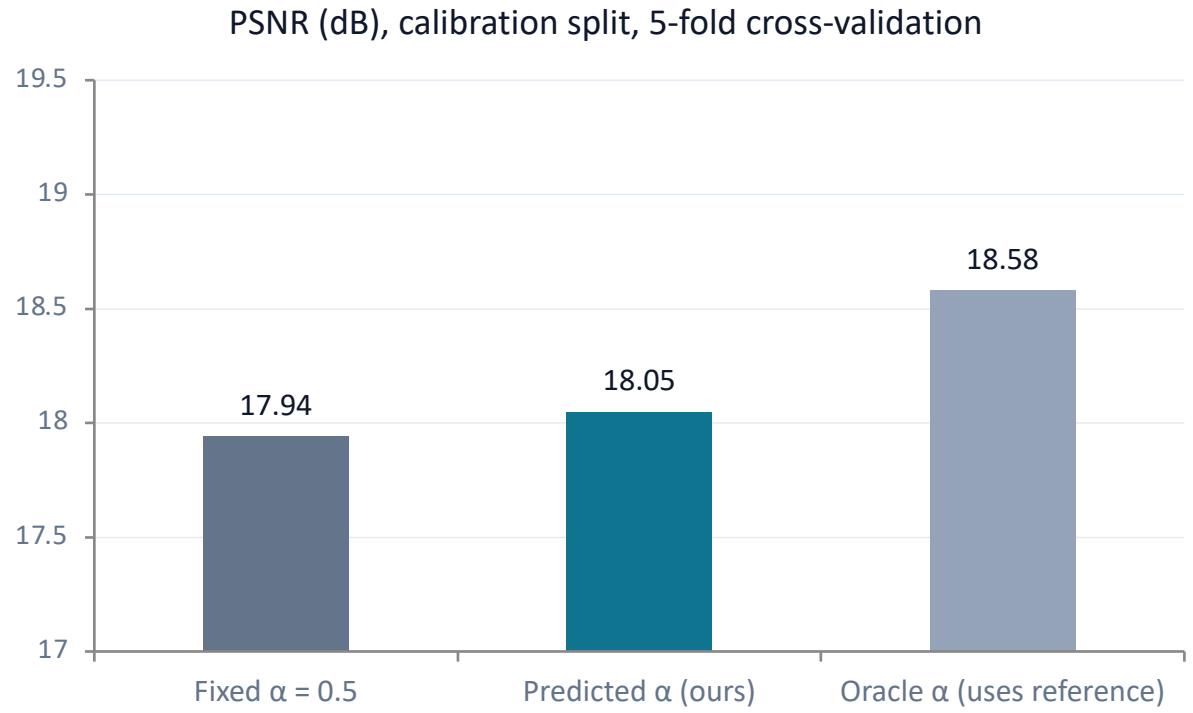
+ removes haze, restores depth
− darker and more saturated

Steps 5–6 · Blending the streams per image



Fused result · predicted α = 0.49 (a roughly even blend)

- α predicted from 26 no-reference statistics
- Per-pixel weights: contrast × saturation × saliency
- 5-level Laplacian pyramid, so no seams



Per-image α has real headroom (+0.64 dB with the oracle). The predictor captures about 17% of it, needing no reference at run time. Axis starts at 17 dB to show the gap.

Evaluation: tune on 30%, report on 70%

268

calibration images · every parameter is chosen here

622

held-out test images · every reported number comes from here

Three calibration stages

Profile multipliers, red-compensation strength, then the Module 7 model — all on the calibration split, scored by PSNR and SSIM.

Fair ablation

Routing off means one global profile tuned with the same grid and data, not untuned defaults.

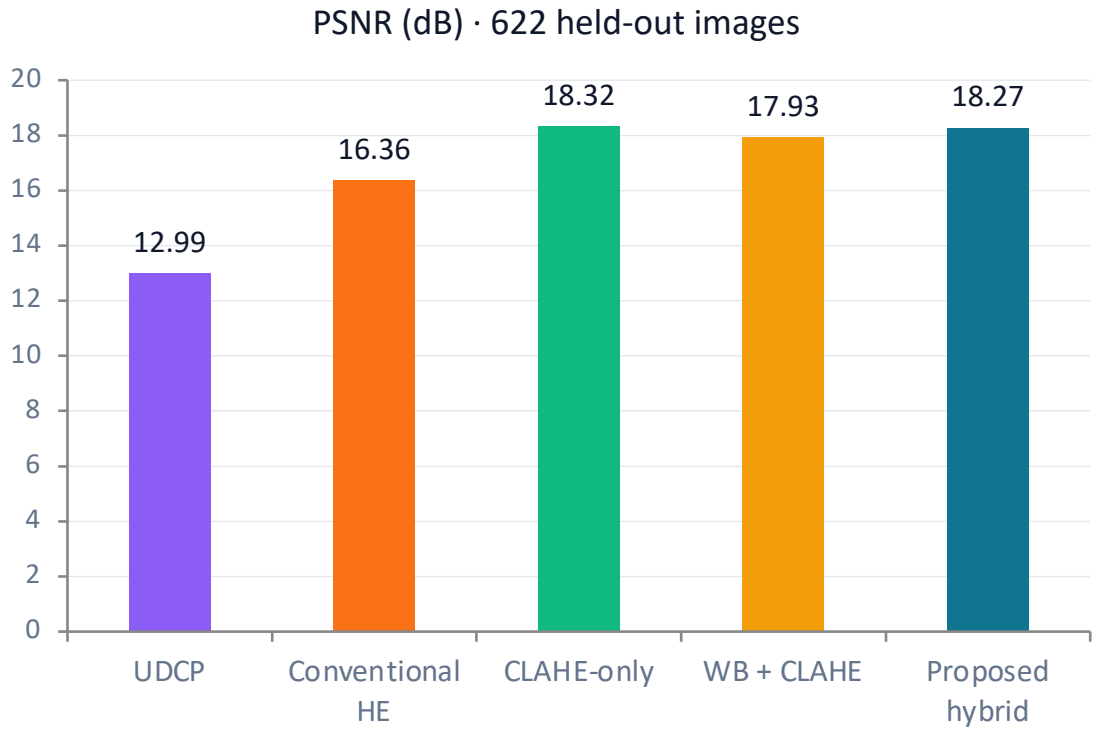
Statistics

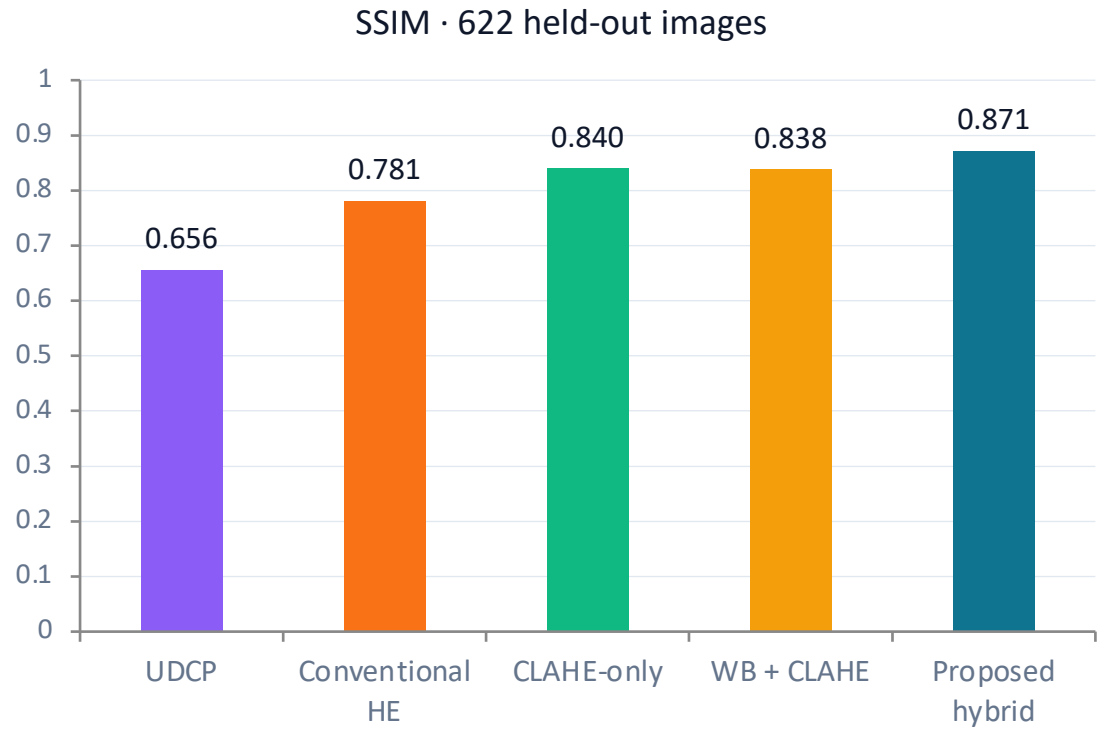
Paired Wilcoxon tests, Holm correction, rank-biserial effect sizes and bootstrap 95% CIs.

Reference caveat

UIEB references are the enhancement volunteers preferred among 12 methods, not ground truth.

Results: best SSIM, clearly ahead of UDCP, HE and WB+CLAHE

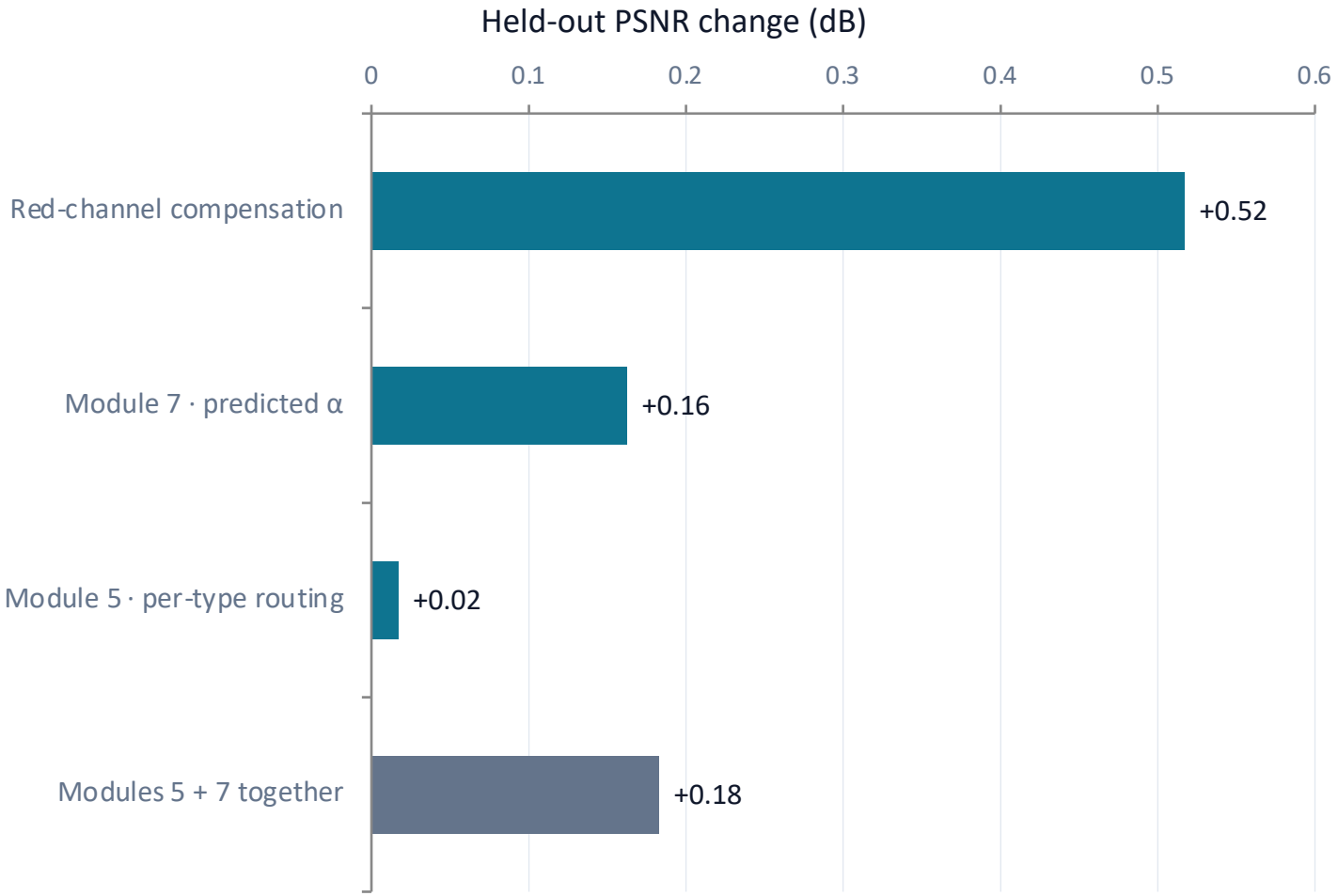




vs CLAHE-only: −0.04 dB PSNR (negligible effect), +0.032 SSIM · **vs HE:** +1.91 dB, +0.091 SSIM · **vs WB+CLAHE:** +0.34 dB, +0.033 SSIM · **vs UDCP:** +5.29 dB, +0.216 SSIM

All differences significant after Holm correction except PSNR vs CLAHE-only. Mean paired differences per image.

What each part contributes



SSIM change

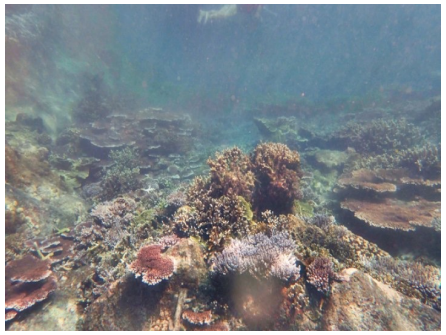
+0.024	Red-channel compensation
+0.003	Module 7 · predicted α
+0.002	Module 5 · per-type routing
+0.004	Modules 5 + 7 together

Red compensation measured on the no-module baseline; Modules 5 and 7 against the equally tuned control.

Visual comparison on held-out images

Raw	Conventional HE	CLAHE-only	Proposed hybrid	UIEB reference
				

green turbid · PSNR / SSIM — proposed 21.59 / 0.932, CLAHE-only 20.91 / 0.921, HE 18.82 / 0.884

hazy bright · PSNR / SSIM — proposed 15.52 / 0.797, CLAHE-only 15.52 / 0.771, HE 14.29 / 0.746

Where it still falls short



Raw



CLAHE-only



Proposed hybrid



UIEB reference

No PSNR gain over CLAHE-only

Level overall (−0.04 dB, negligible, not significant). CLAHE-only scores higher PSNR on 45% of test images, like this clear blue one (17.13 vs 19.24 dB).

Routing adds little

A per-image oracle profile would add 1.45 dB over one global profile; four water types capture +0.02 dB of it.

Calibration at the grid edge

WB cap 1.5, CLAHE × 0.6 and ω 0.5 are the mildest values searched: the references prefer gentler processing than the grid allowed.

Rigour: issues we found and fixed

UCIQE on 8-bit Lab

OpenCV's encoding made grey pixels read as chroma ≈181; now computed in true CIELAB.

Keypoint cap tied counts

ORB stopped at 1,000 points, so raw and restored images hit the same ceiling; cap raised to 200,000.

White-balance cap unused

The per-type gain cap was defined but never applied; now passed through.

Tuned on the test images

Hand-set profiles were judged on the same images; now a held-out 30/70 split.

No multiple-comparison control

25 raw p-values; now Holm correction, effect sizes and 95% CIs.

Gamma convention inverted

γ was derived for input^γ but applied as input^(1/γ), inverting the adaptivity; fixed and re-calibrated.

Live demo

1. Compare tab: pick 773_img_ (green turbid) and compare against every method
2. Pipeline tab: walk the stages and drag the α preview slider
3. Toggle Module 5 and Module 7 off and on, watching PSNR/SSIM in the header
4. Metrics tab: no-reference scores vs. the PSNR/SSIM table
5. Downstream tab: run ORB keypoint and edge validation
6. Switch to 699_img_ (clear blue) to show the limitation



Streamlit app

```
streamlit run
app/streamlit_app.py
```

Every metric is computed live on the selected image

Takeaways

Red-channel compensation before white balance is the largest single gain: +0.52 dB PSNR on held-out images.

The full hybrid has the highest SSIM of all compared methods (0.871), matches CLAHE-only on PSNR, and clearly beats UDCP, HE and WB+CLAHE on 622 held-out images.

A calibrated per-image fusion weight adds +0.16 dB; every parameter was tuned on 268 images and every claim tested on the other 622.

Thank you · Questions?